

Overview

In this project named School Library, we develop a website that might be used to manage the catalog for a local library.

In this project we will use the following:

- Use Django's tools to create a skeleton website and application.
- Create models to represent your application's data.
- Use the Django admin site to populate your site's data.
- Create views to retrieve specific data in response to different requests, and templates to render the data as HTML to be displayed in the browser.
- Create mappers to associate different URL patterns with specific views.
- Add user authorization and sessions to control site behavior and access.
- Work with forms.

Project's folder/file structure

main website folders/files:

```
shcoollibrary/  
  manage.py  
  catalog  
  schoollibrary/  
    __init__.py  
    settings.py  
    urls.py  
    wsgi.py  
    asgi.py  
  templates  
  .env  
  mysql.cnf
```

The *schoollibrary* project sub-folder is the entry point for the website:

- **__init__.py** is an empty file that instructs Python to treat this directory as a Python package.
- **settings.py** contains all the website settings, including registering any applications we create, the location of our static files, database configuration details, etc.
- **urls.py** defines the site URL-to-view mappings. While this could contain *all* the URL mapping code, it is more common to delegate some of the mappings to particular applications, as you'll see later.

- **wsgi.py** is used to help your Django application communicate with the web server. You can treat this as boilerplate.
- **asgi.py** is a standard for Python asynchronous web apps and servers to communicate with each other. Asynchronous Server Gateway Interface (ASGI) is the asynchronous successor to Web Server Gateway Interface (WSGI). ASGI provides a standard for both asynchronous and synchronous Python apps, whereas WSGI provided a standard for synchronous apps only. ASGI is backward-compatible with WSGI and supports multiple servers and application frameworks.
- **templates** is a folder for additional html templates and required for user login, logout etc.
- **.env** is a file for storing sensitive secret key for generating hashes.
- **mysql.cnf** is a file for storing database connection's configurations.
- **catalog** is our main webapp directory.

The **manage.py** script is used to create applications, work with databases, and start the development web server.

catalog/ sub folders/files:

```
catalog/  
    admin.py  
    apps.py  
    models.py  
    urls.py  
    forms.py  
    views.py  
    __init__.py  
    static  
    templates  
    migrations
```

A *migrations* folder, used to store "migrations" — files that allow you to automatically update your database as you modify your models.

- **__init__.py** — an empty file created here so that Django/Python will recognize the folder as a [Python Package](#) and allow you to use its objects within other parts of the project.
- **apps.py** store our app configuration like name etc for registering app in site.
- **models.py** used to define database structure using classes as table definition(called models)and some more database related feature.

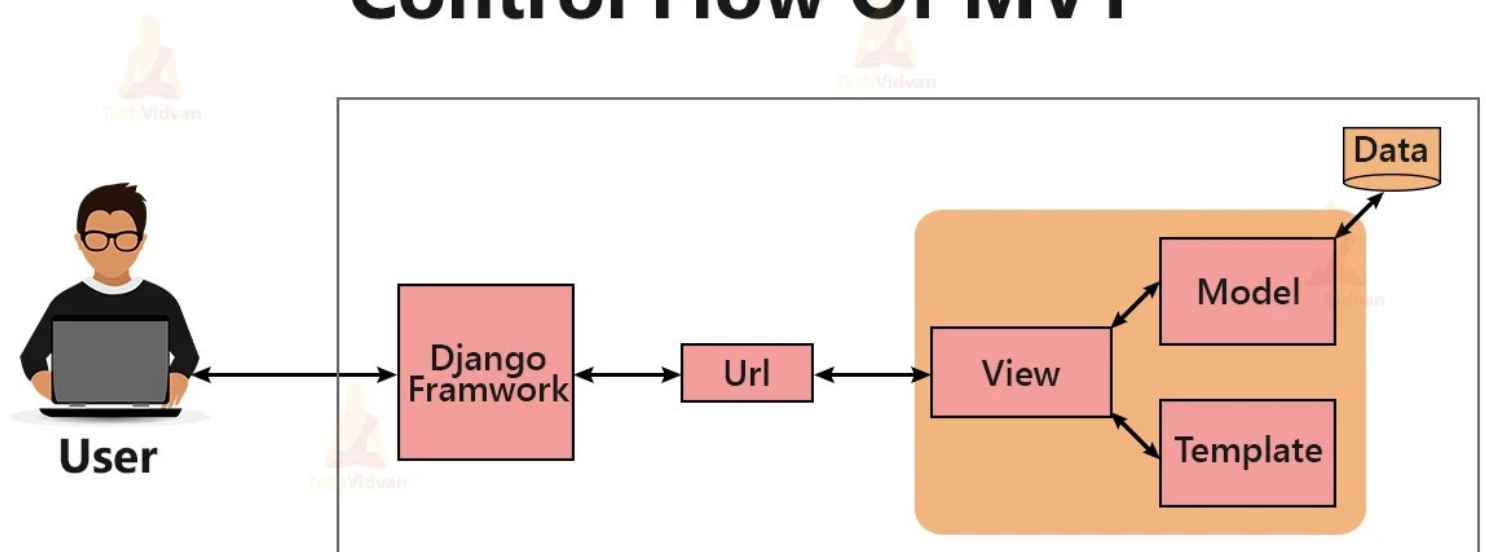
- **urls.py** used to map our catalog app related urls.
- **admin.py** is used for registering database's models ,so admin of site can see, modify and delete data from database in default administration dashboard.
- **forms.py** is used for managing and building custom forms.
- **views.py** - in this file we write views which handles requests for specific url and return response. These views are mapped with urls in urls.py file.
- **static** folder store static files such as image,css etc.
- **templates** this folder store html files of our catalog app.

Website's Mechanism

our website's mechanism divided into three main part:

model, view and template.

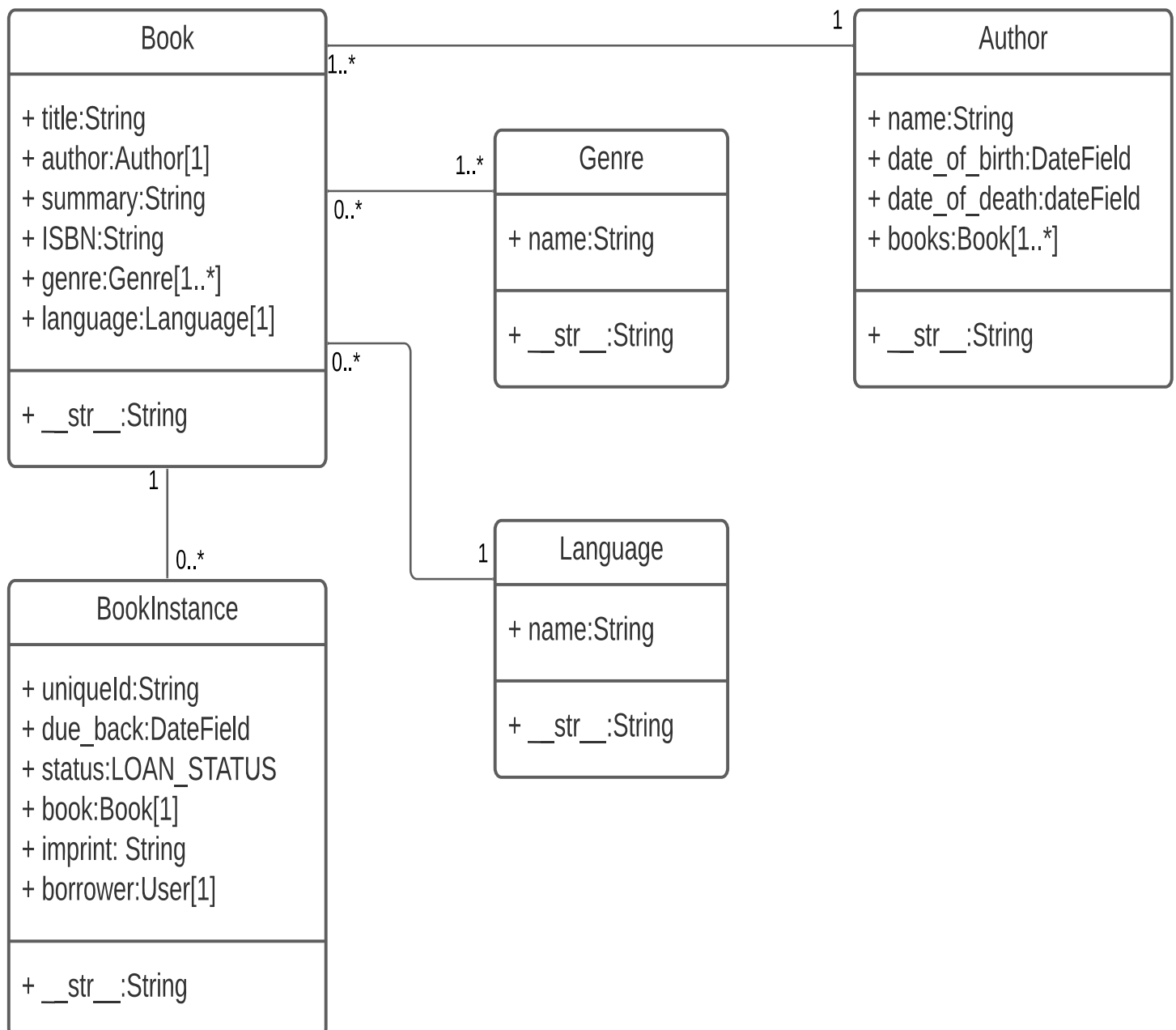
Control Flow Of MVT



Models of website's database

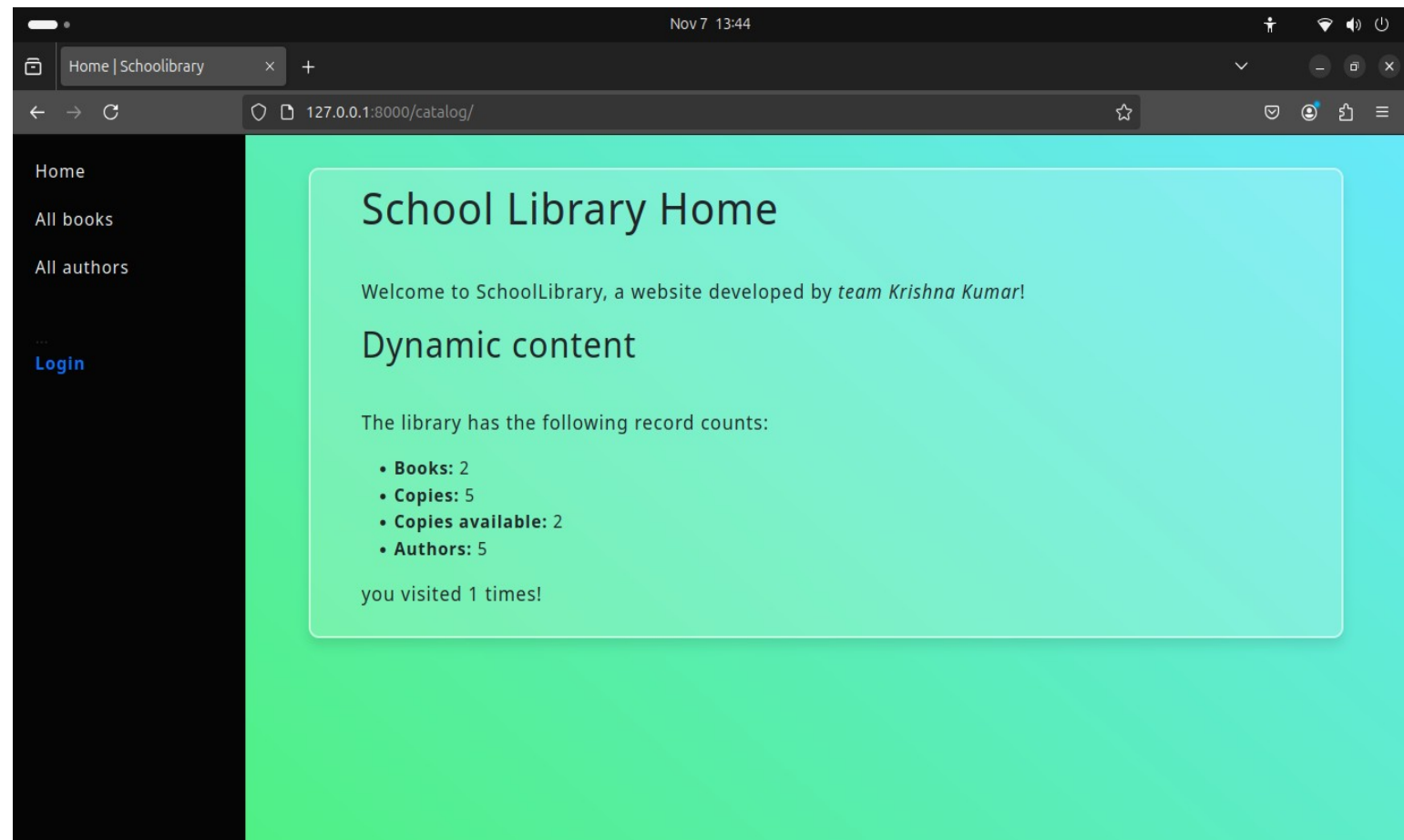
we have models for managing data of book, author, genre, book instance and language.
for users data we used django's default model.

diagram of model structures and relationships



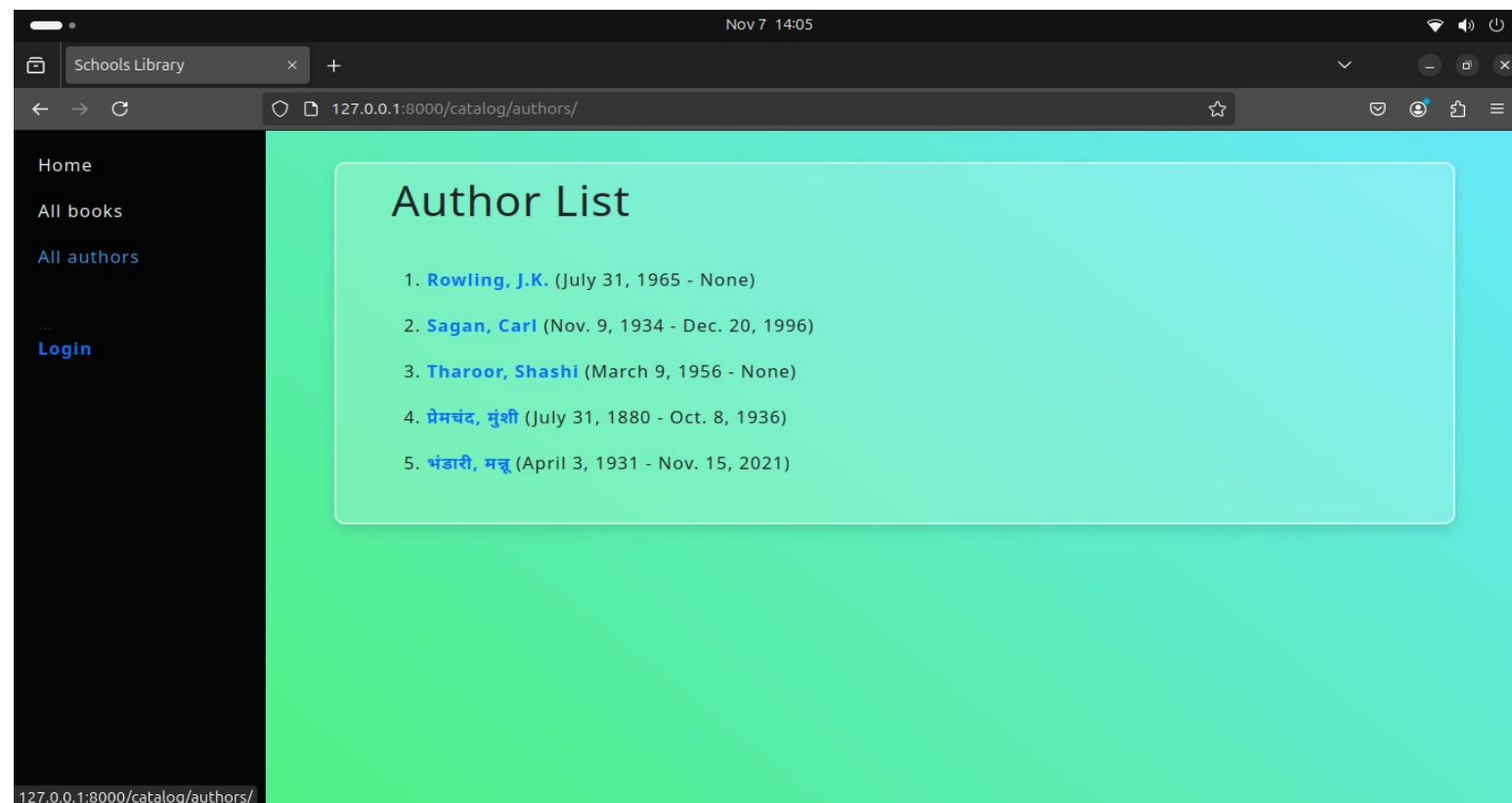
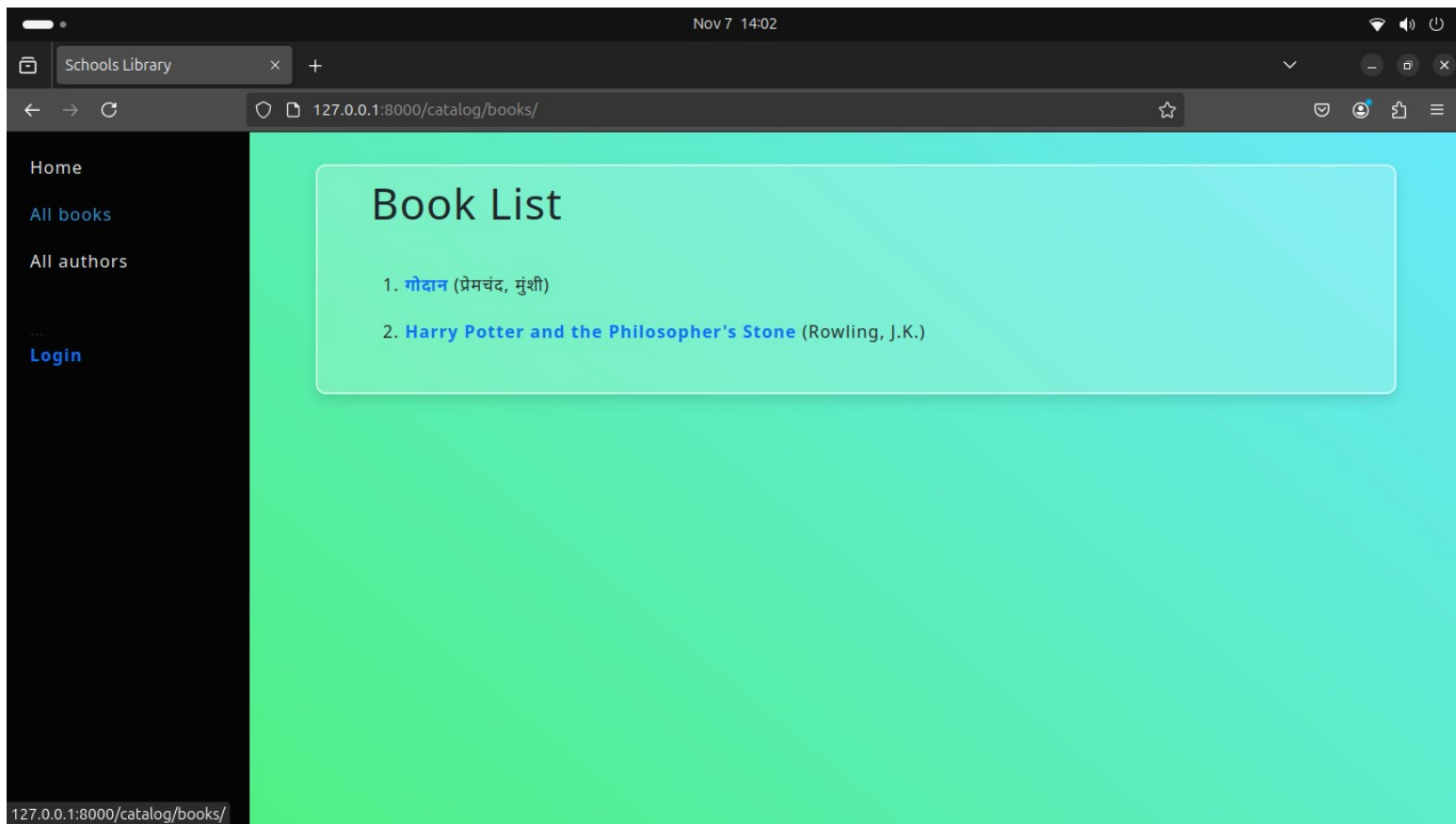
Website Features & Functionality

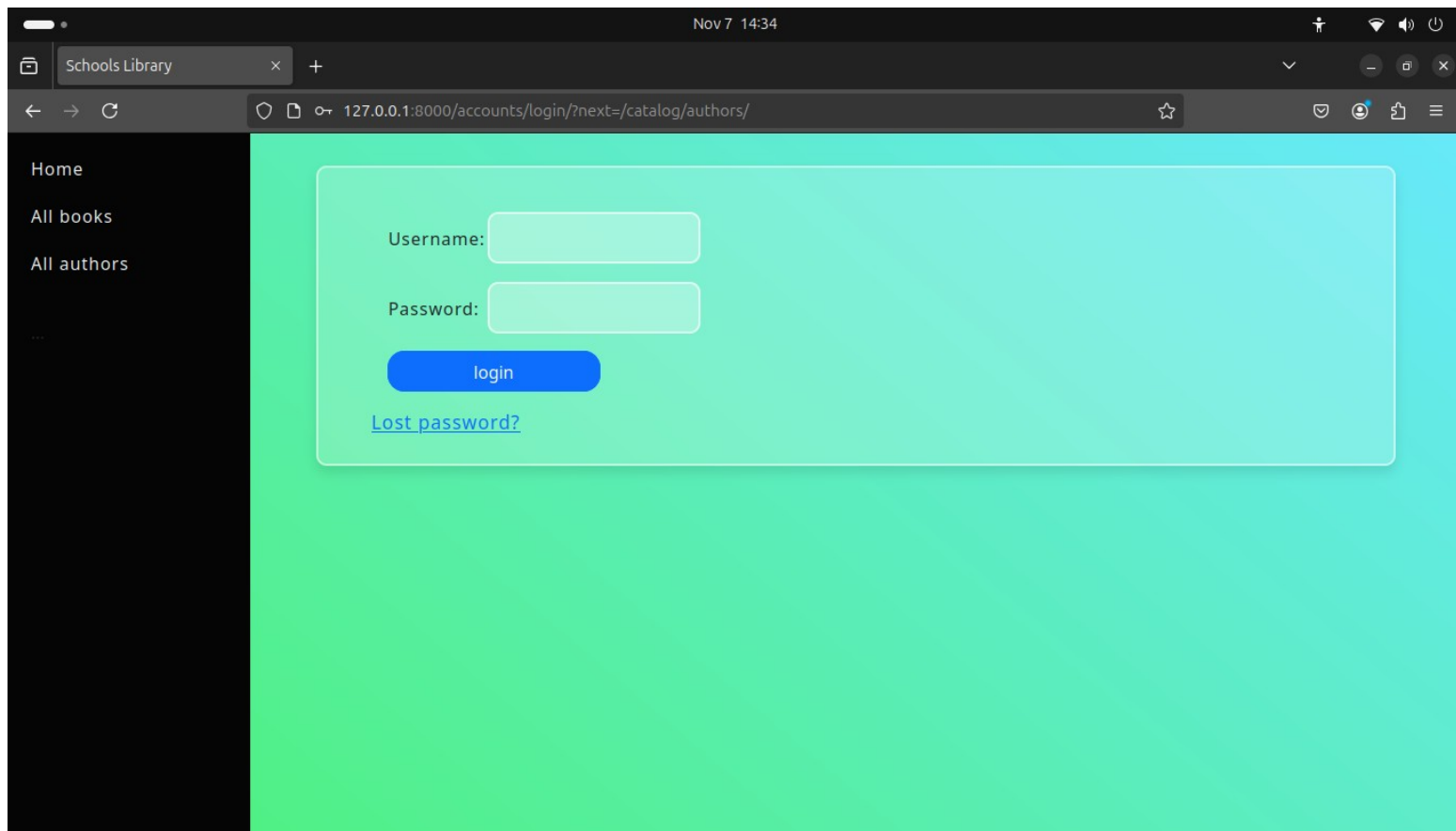
this our home page:



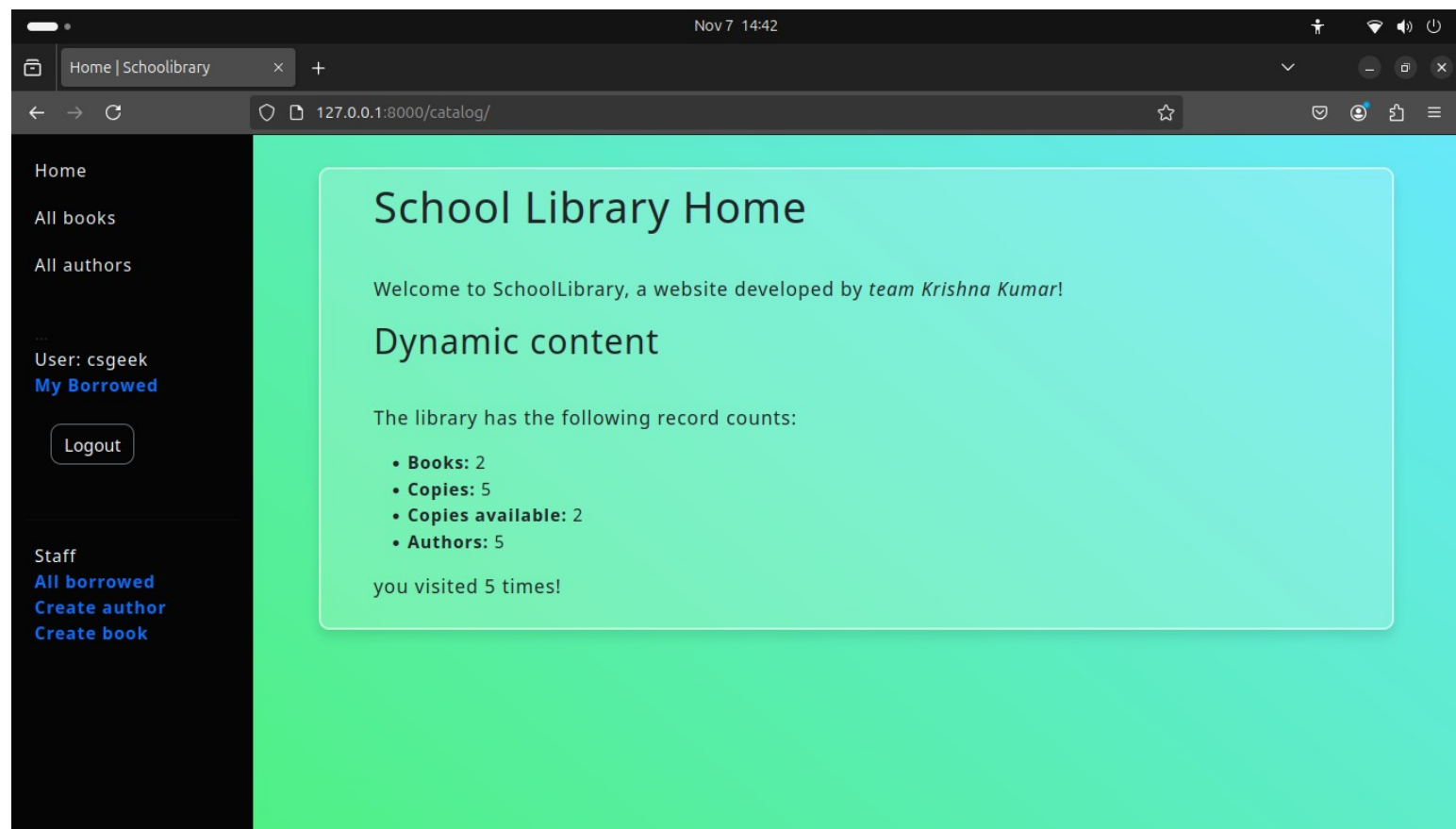
- we can see books by clicking All books link.
- we can see authors by clicking All authors link.
- we can return to main page by clicking Home link.
- we can login into the site by clicking login link.

The pages of link following above given below:



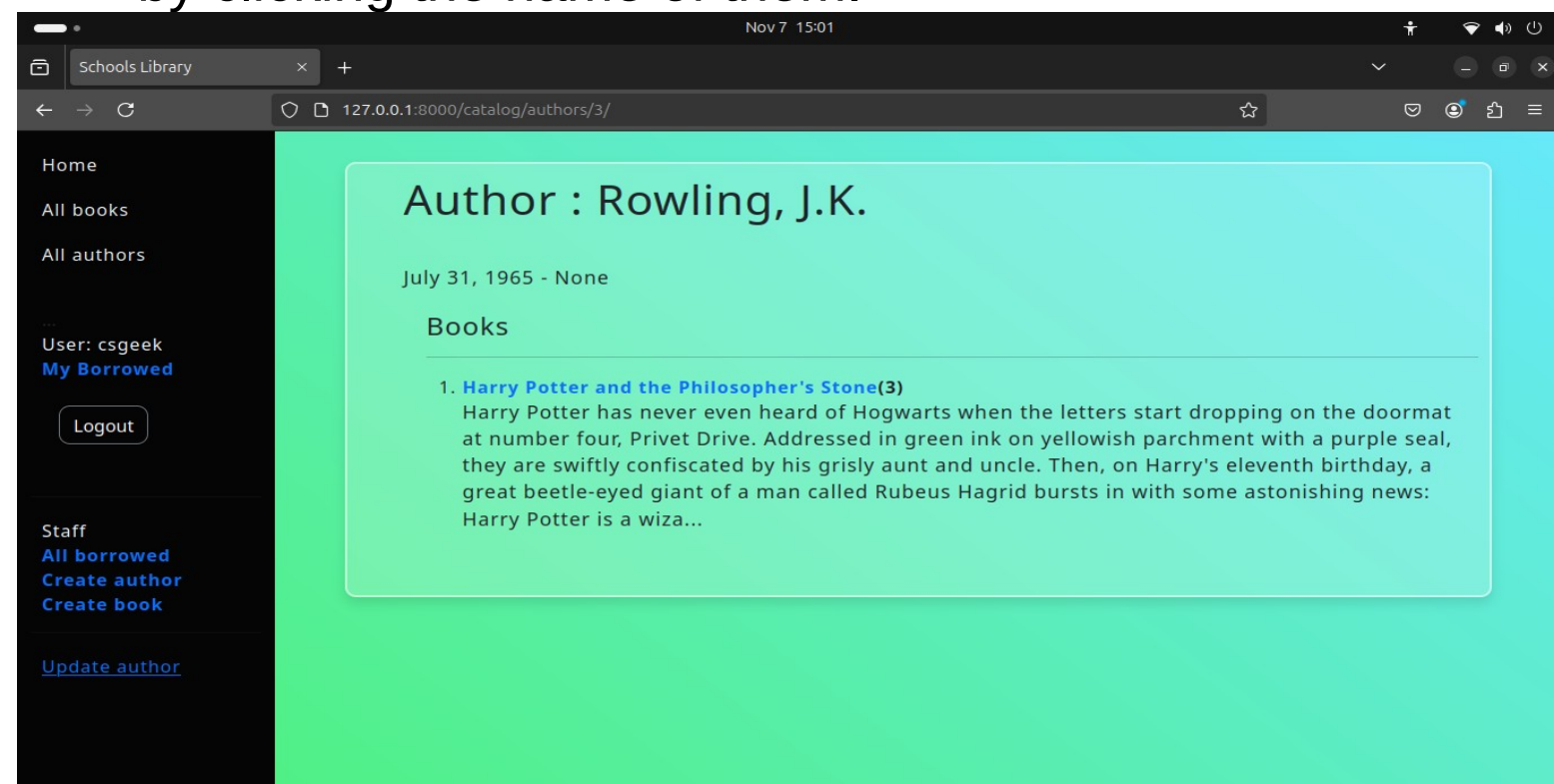


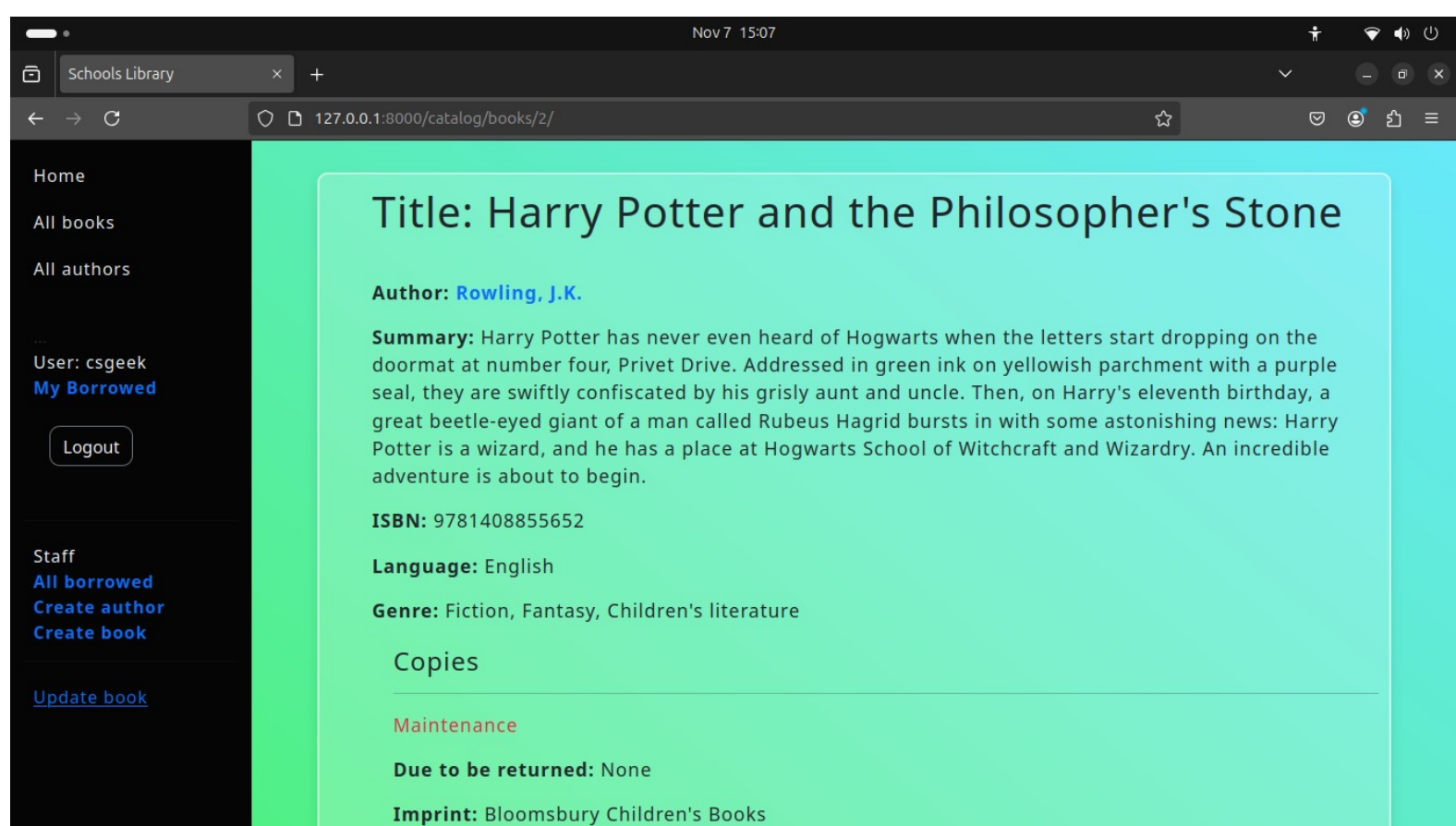
after login as a staff member user:



we can see links which are accessible to staff.

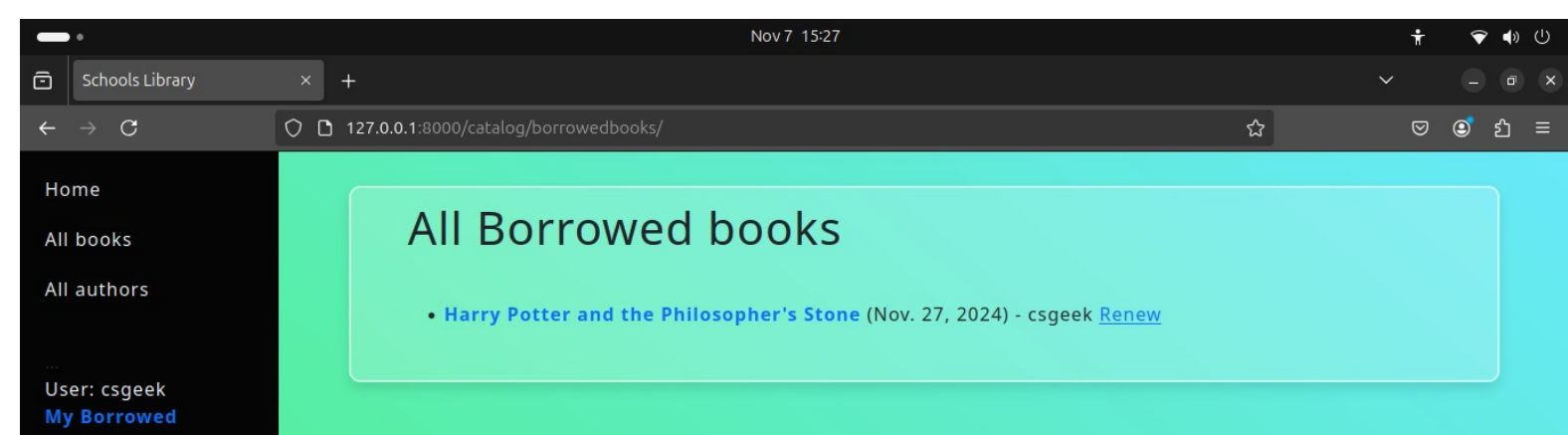
now we can see detail view of an author or a book by clicking the name of them.





user can see his borrowed book's copies by clicking My Borrowed.

staff member can see borrowed book's copies of any user by clicking All borrowed and and can renew the due back date:



renew due back date form:

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:8000/catalog/book/c82dd8f6-490e-403c-9db6-2d26b1551cbf/renew/'. The page has a dark sidebar on the left with links: Home, All books, All authors, User: csgeek, My Borrowed, Logout, Staff, All borrowed, and Create author. The main content area has a light blue background and contains a form titled 'Renew: Harry Potter and the Philosopher's Stone'. The form displays 'Borrower: csgeek' and 'Due date: Nov. 27, 2024'. It includes a 'Renewal date' field with the value '2024-11-28' and a note 'Enter a date between now and 4 weeks (default 3)'. A blue 'Submit' button is at the bottom of the form.

Home

All books

All authors

User: csgeek

[My Borrowed](#)

[Logout](#)

Staff

[All borrowed](#)

[Create author](#)

Renew: Harry Potter and the Philosopher's Stone

Borrower: csgeek

Due date: Nov. 27, 2024

Renewal date: 2024-11-28

Enter a date between now and 4 weeks (default 3).

[Submit](#)

Creation Deletion and Updation of Author and Book:

note :- only staff member can do these operations.

- to create author click the Create Author link, which opens the form as given in the picture.

The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8000/catalog/author/create/`. The page has a dark sidebar on the left with links: Home, All books, All authors, User: csgeek, My Borrowed, Logout, and Staff. The main content area has a light blue background and contains a form with the following fields: First name (empty), Last name (empty), Date of birth (empty), and Died (11/11/2023). A blue Submit button is at the bottom of the form.

First name:

Last name:

Date of birth:

Died:

- to update author first go to author detail view and then click the generated link Update author then a form opens given in the picture. Now you can change the values.

The screenshot shows a web browser window with the address bar displaying `127.0.0.1:8000/catalog/author/3/update/`. The page has a dark sidebar on the left with links: Home, All books, All authors, User: csgeek, My Borrowed, Logout, and Staff. The main content area has a light blue background and contains a form with the following fields: First name (J.K.), Last name (Rowling), Date of birth (1965-07-31), and Died (empty). A blue Submit button is at the bottom of the form.

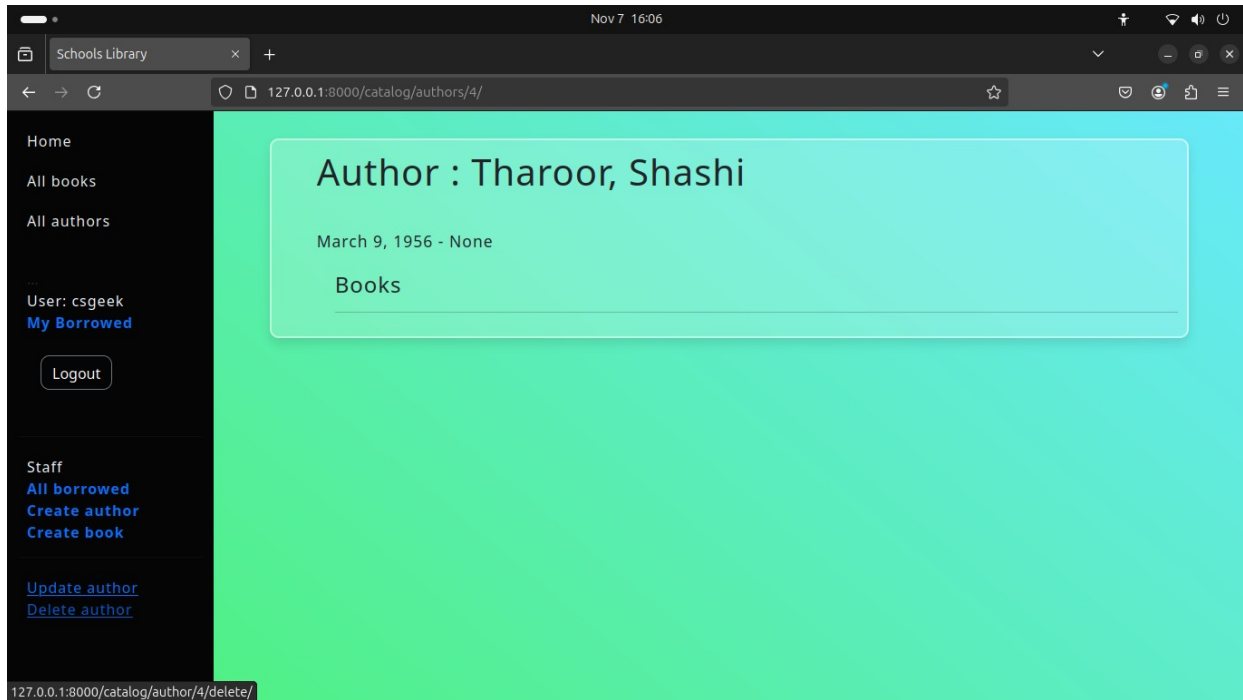
First name:

Last name:

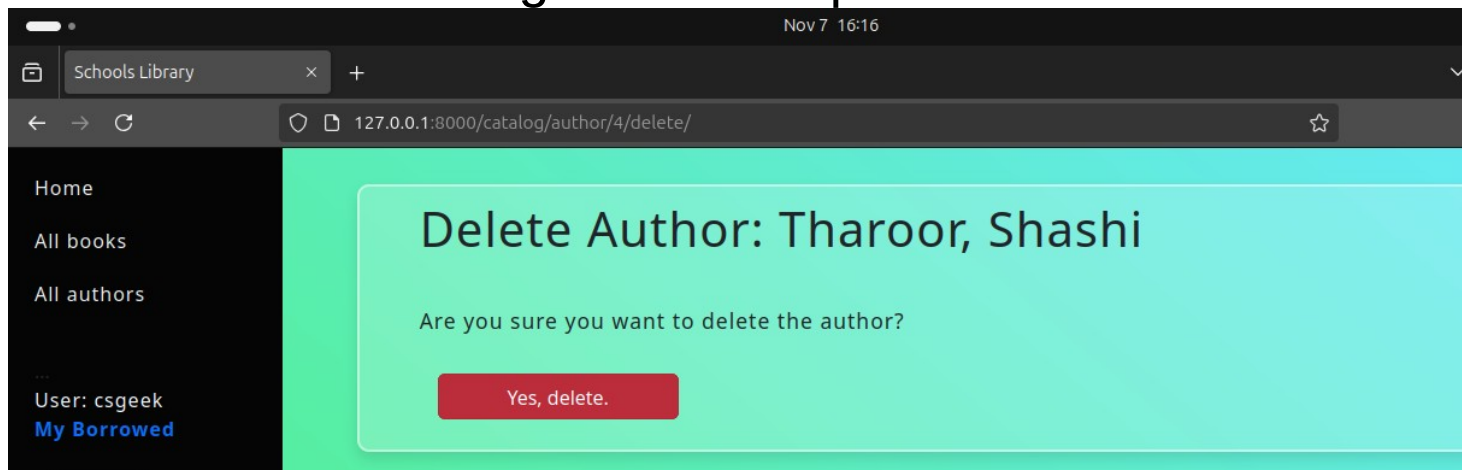
Date of birth:

Died:

- we can only delete the author who's books are not in the record.



- now click the Delete author link then form ask confirmation click the confirmation button as shown in the given below picture.



- to create book click create book link and fill the form, shown in the given below picture.

Nov 7 16:23

Schools Library

127.0.0.1:8000/catalog/book/create/

Home

All books

All authors

User: csgeek

My Borrowed

Logout

Staff

All borrowed

Create author

Create book

Title:

Author:

Summary:

Enter a brief description of the book

ISBN:

13 Character [ISBN number](#)

Genre:

Children's literature

Fantasy

Fiction

Select a genre for this book

Language:

Select a language for this book

Submit

127.0.0.1:8000/catalog/book/create/

- to update book, which is similar to author update given above. Find the name of book, click update book link and fill the form, shown in the given below picture.

Nov 7 16:31

Schools Library

127.0.0.1:8000/catalog/book/2/update/

Home

All books

All authors

User: csgeek

My Borrowed

Logout

Staff

All borrowed

Create author

Create book

Title: Harry Potter and the Ph

Author: Rowling, J.K. ▾

Summary: Harry Potter has never even heard of Hogwarts when the letters start dropping on the doormat at number four, Privet Drive. Addressed in green ink on yellowish parchment with a purple seal, they are swiftly confiscated by his grisly aunt and uncle. Then, on Harry's eleventh birthday, a great beetle-eyed giant of a man called Rubeus Hagrid bursts in with some astonishing news: Harry Potter is a wizard, and he has a place at Hogwarts School of Witchcraft and Wizardry. An incredible adventure is about to begin.

Enter a brief description of the book

ISBN: 9781408855652

13 Character [ISBN number](#)

Genre: Children's literature

Fantasy

Fiction

Select a genre for this book

Language: English ▾

Select a language for this book

Submit

- deletion of book is similiar deletion of author,just click the name of book ,if book have no copy only then you can see delete link and then you can delete it.